

Tangent Error

The earth rotates a full 360 degrees in around 86164s (approx. 24 hours) in a day.

This means that the stars and other celestial bodies in the night sky appear to be moving at an angular speed of:

$$\omega = \frac{2\pi}{86164.0905}$$

This is the same angular speed at which my star tracker should move.

However, the rod doesn't directly rotate the camera, instead it simply pushes the two boards apart, which follow a linear path instead of an arc.

Unless mechanical or speed-related changes are made (e.g bending the rod or altering the speed of separation) this will result in tangent error.

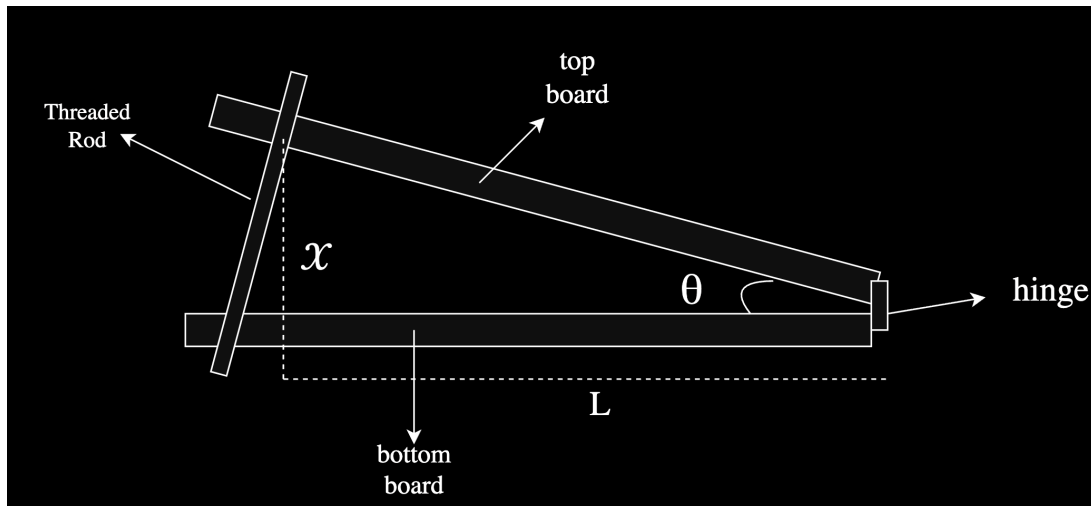
Optimum constant speed

In the first iteration of my experiment, the speed of the motor will remain constant and will not account for tangent error. The objective here is to find the optimum linear speed at which the two boards should separate to minimise the disadvantage of having a constant speed. This calculation can prove helpful for users who plan on doing short exposure photography at short focal lengths, where tangent error is negligible.

The Math for an Ideal scenario

Lets suppose

- L = distance from the hinge to rod
- x = rod extension
- θ = angle of rotation of the camera



Using basic trigonometry we can deduce ;

$$x = L \tan(\theta)$$

The angle must increase linearly with time as;

$$\theta = \omega t$$

Combining these two equations gives us the ideal rod position as a function of time.

$$x(t) = L \tan(\omega t)$$

You may notice that this is not a straight line because the tangent function curves upward. This proves that we cannot use a constant speed setup to ideally match the rotation of the earth, Therefore, it is necessary to determine the constant speed that provides the closest approximation over the desired exposure time.

Determining the Optimum Constant Speed

There are several possible methods for selecting the constant speed of a straight-rod barn door tracker, each with its own advantages and limitations. The simplest approach is to match the initial speed of the ideal motion. Since the ideal board separation is given by

$$x(t) = L \tan(\omega t),$$

the initial speed is obtained by differentiating this expression and evaluating it at $t = 0$

$$\left. \frac{dx}{dt} \right|_{t=0} = L\omega .$$

This method provides a perfect match only at the beginning of the exposure. As time progresses, the tangent function curves upward while the motor continues to move at a constant linear speed. Consequently, the tracking error increases with time, making this approach unsuitable for longer exposures.

CHOICE OF APPROXIMATION METHOD

To determine the optimum constant speed, this investigation adopts the minimax approximation (also known as uniform approximation).

Minimax approximation is a mathematical technique that approximates a complex function with a simpler one by minimising the **maximum absolute error** over a specified interval. Rather than reducing the average error, it seeks to make the largest deviation between the approximation and the true function as small as possible.

Mathematically, the minimax condition is

$$\left| E_{\max} \right| = \left| E_{\min} \right| ,$$

Or equivalently,

$$E(T) = - E(t_c),$$

Where

- T is the exposure duration,

- t_c

is the time at which the maximum positive angular error occurs.

This condition ensures that the largest positive and negative tracking errors have equal magnitude, producing the smallest possible worst-case error over the entire exposure.

WHY MINIMAX IS APPROPRIATE

In astrophotography, image quality is limited by the **largest tracking error**, not the average tracking error. Even a single pixel of star trailing can noticeably reduce image sharpness. Therefore, reducing the worst-case tracking error is considerably more important than reducing numerous small errors throughout the exposure.

For this reason, the minimax criterion is particularly well suited to the problem. By balancing the maximum positive and negative angular errors, it produces the constant motor speed that minimises the greatest possible deviation from ideal sidereal tracking during the chosen exposure.

Defining the Error Function

The ideal angular position of the tracker after time t is

$$\theta_{\text{ideal}} = \omega t,$$

where

$$\omega = \frac{2\pi}{86164.0905}.$$

The actual angle produced by the tracker is determined by the geometry of the barn door mechanism. Since the board separation is

$$x = vt,$$

the actual angle is

$$\theta_{\text{actual}} = \arctan\left(\frac{vt}{L}\right),$$

where

- v is the constant linear speed
- L is the hinge to threaded rod distance

The angular tracking error is therefore defined as the difference between the actual and ideal angular positions,

$$E(t) = \theta_{\text{actual}} - \theta_{\text{ideal}} = \arctan\left(\frac{vt}{L}\right) - \omega t.$$

This function describes how far ahead or behind the tracker is relative to the ideal sidereal motion at any instant during the exposure. Positive error will indicate how much the tracker has rotated **away** from the sidereal position and negative error will indicate how much it is lagging **behind**.

Since the objective is to minimise the worst-case tracking error, the locations of the maximum positive and maximum negative errors must be determined.

Finding the Maximum and Minimum Error

The largest errors will be found upon differentiating the error function in order to find the stationary point(s):

$$E'(t) = \frac{\frac{v}{L}}{1 + \left(\frac{vt}{L}\right)^2} - \omega.$$

$$E'(t) = 0.$$

$$\frac{\frac{v}{L}}{1 + \left(\frac{vt}{L}\right)^2} = \omega.$$

Solving for t gives

$$t_c = \frac{L}{v} \sqrt{\frac{v}{L\omega} - 1}.$$

To determine whether this is the maximum or the minimum we must find the second derivative which is

$$E''(t) = - \frac{2v^3 t / L^3}{\left(1 + \left(\frac{vt}{L}\right)^2\right)^2}.$$

As $v > 0$, $L > 0$ and $t > 0$, we can safely say that:

$$E''(t) < 0$$

For all positive values of t, the stationary point is always a **maximum**.

Furthermore, because the second derivative never changes sign, the error function has **only one stationary point**. For the 20-minute exposure considered

in this investigation, this stationary point corresponds to the maximum positive error, while the maximum negative error occurs at the end of the exposure.

Note.

The analysis presented in this section assumes that the stationary point of the error function lies within the chosen exposure interval, that is,

$$0 < t_c < T$$

Under this condition, the error function has one interior maximum, while the maximum negative error occurs at the end of the exposure

For different combinations of hinge distance and exposure time, however, the stationary point may occur after the exposure has ended. In this case, the error function has no interior maximum during the exposure, and the maximum negative error does not necessarily occur at the endpoint.

Therefore, the derivation presented here is specific to the 20-minute exposure investigated in this report and should not be assumed to apply universally.

To accommodate all possible exposure times and hinge lengths, the calculator developed as part of this investigation first determines whether an interior stationary point exists within the exposure interval. It then applies the appropriate optimisation procedure for that particular case.

Applying the Minimax Criterion

Having identified the maximum positive and maximum negative angular errors, the optimum constant linear speed can now be determined using the minimax criterion.

The minimax approximation seeks to minimise the largest absolute tracking error over the exposure interval. This occurs when the maximum positive and maximum negative errors are equal in magnitude and opposite in sign.

Therefore, the optimum speed satisfies

$$E(t_c) = -E(T).$$

Substituting the previously found error function into this minimax condition gives

$$\arctan\left(\frac{vt_c}{L}\right) - \omega t_c = -\left(\arctan\left(\frac{vT}{L}\right) - \omega T\right).$$

The aforementioned time of the stationary point is

$$t_c = \frac{L}{v} \sqrt{\frac{v}{L\omega} - 1},$$

The final equation we get

$$\arctan\left(\frac{v}{L} \left(\frac{L}{v} \sqrt{\frac{v}{L\omega} - 1}\right)\right) - \omega \left(\frac{L}{v} \sqrt{\frac{v}{L\omega} - 1}\right) = -\left(\arctan\left(\frac{vT}{L}\right) - \omega T\right).$$

These equations completely define the optimum speed. However, they cannot be rearranged to obtain an explicit formula for v . Therefore, a numerical optimisation algorithm is used to determine the value of v that satisfies the minimax criterion. Which can be used by clicking [here](#).